The Fokker–Planck Equation

Derivation and Applications in Economics and Finance

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The Fokker–Planck (Kolmogorov forward) equation governs how the probability density of a diffusion process changes over time. This note derives it, first for Brownian motion and then for a general stochastic differential equation, and works out its solutions for three processes that recur in economics and finance: the Ornstein–Uhlenbeck process, geometric Brownian motion, and the Cox–Ingersoll–Ross process, whose stationary law shows where the Feller condition comes from. It then describes two uses of the equation in finance and economics, recovering local volatility from option prices and tracking wealth distributions in heterogeneous-agent models, and surveys grid-based and learning-based methods for solving it numerically.

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1 Background

The Fokker–Planck equation (FPE) describes how the probability density of a continuous-time Markov diffusion evolves. When a quantity follows a stochastic differential equation (SDE), the SDE describes individual sample paths, while the FPE describes the distribution of those paths: how likely the process is to be in each region of the state space at each date. The two are equivalent descriptions of the same process (Risken 1989; Gardiner 2009). Simulating the SDE is often the practical choice in many dimensions. The FPE is the natural tool when the object of interest is the density itself, such as a stationary distribution or the probability of reaching a boundary.

The equation grew out of the study of Brownian motion, the erratic movement of a small particle pushed around by the molecules of a surrounding fluid. Fokker (1914) derived it in a study of the average energy of rotating electric dipoles in a radiation field, and Planck (1917) extended it to the statistical behavior of large ensembles of particles. Kolmogorov (1931) obtained the same equation within his general analytic theory of Markov processes, which is why it is also called the Kolmogorov forward equation. Because it describes how a population of randomly moving units spreads out, the equation has found uses far from physics, including in finance and economics.

1.1 Applications in finance

The Black–Scholes model assumes that an asset's volatility is constant. Option prices contradict this assumption. In the words of Dupire (1993), "Black-Scholes volatilities implied from market prices exhibit a strike pattern, commonly termed 'Smile', as well as a term structure." No single constant volatility reproduces all observed option prices at once.

Dupire (1994) showed that the forward equation resolves this within a one-factor diffusion model. Suppose that under the risk-neutral measure the asset price follows

$$dS_t = (r - q)\, S_t\, dt + \sigma(S_t, t)\, S_t\, dW_t,$$

with interest rate r and dividend yield q. The risk-neutral density $p(K, T)$ of S_T then satisfies the FPE in the variables (K, T), and Breeden and Litzenberger (1978) showed that this density can be read off call prices: $\partial^2 C/\partial K^2 = e^{-rT} p(K, T)$. Substituting this relation into the FPE and integrating twice in K gives Dupire's formula for the local volatility that reproduces every observed call price:

$$\sigma^2(K, T) = \frac{\partial_T C + (r - q)\, K\, \partial_K C + q\, C}{\frac{1}{2} K^2\, \partial_{KK} C}. \tag{1}$$

Brigo and Mercurio (2002) use the same equation in the opposite direction. They choose the local volatility so that the solution of the FPE is a mixture of lognormal densities, which generates smiles while keeping option prices in closed form (see also Brigo et al. 2003).

1.2 Applications in economics

In economics the FPE describes how a distribution across many agents evolves. In kinetic models of wealth exchange, an FPE arises as the limit of many small trades, and its stationary solutions can have the heavy upper tails seen in wealth data (Furioli et al. 2017). Heavy, power-law tails are a recurring empirical regularity, appearing in the distributions of income, wealth, and the sizes of firms and cities (Achdou et al. 2014).

The most widely used application today is in continuous-time models with heterogeneous agents (Achdou et al. 2022). Households' optimal saving is characterized by a Hamilton–Jacobi–Bellman (HJB) equation, and the cross-sectional distribution of wealth and income evolves according to a Kolmogorov forward equation driven by that saving behavior. Equilibrium prices, such as the interest rate, link the two equations. This backward–forward system is what Lasry and Lions (2007) call a mean field game, and it has made the FPE a standard tool in quantitative macroeconomics.

The FPE also suits a broader view of the economy as a complex adaptive system, in which agents continually adjust their behavior in response to the aggregate patterns that their own behavior creates (Arthur 2013). A density-level description is a natural language for such aggregate patterns, because it tracks how a whole population is distributed rather than the path of any single agent.

2 Derivation

The derivation has three steps: apply Itô's formula to a smooth test function of the process, take expectations, and integrate by parts to move the derivatives from the test function onto the density.

We do this first for Brownian motion and then for a general SDE. The argument follows the heuristic derivation in Rouah (n.d.); Øksendal (2003) and Pavliotis (2014) give rigorous treatments.

2.1 Brownian motion

Let W_t be a standard Brownian motion: $W_0 = 0$, with independent increments and $W_t - W_s \sim N(0, t-s)$ for $s < t$. Write $p(x,t)$ for the density of W_t, and take a test function $f \in C_c^\infty(\mathbb{R})$, that is, infinitely differentiable and equal to zero outside some bounded interval $[x_1, x_2]$.

By Itô's formula, using $(dW_t)^2 = dt$,

$$df(W_t) = f'(W_t)\, dW_t + \tfrac{1}{2} f''(W_t)\, dt. \tag{2}$$

Integrating from 0 to t and taking expectations gives

$$\mathbb{E}[f(W_t)] - f(0) = \frac{1}{2} \int_0^t \mathbb{E}[f''(W_s)]\, ds,$$

because the Itô integral $\int_0^t f'(W_s)\, dW_s$ has mean zero: its integrand is bounded and adapted. The map $s \mapsto \mathbb{E}[f''(W_s)]$ is continuous by dominated convergence, since f'' is bounded (Rudin 1976, Theorem 11.32), so differentiating in t yields

$$\frac{d}{dt} \int_{-\infty}^{\infty} f(x)\, p(x,t)\, dx = \frac{1}{2} \int_{-\infty}^{\infty} f''(x)\, p(x,t)\, dx. \tag{3}$$

Now integrate the right-hand side by parts twice. Because f and f' vanish outside $[x_1, x_2]$, every boundary term is zero:

$$\int_{-\infty}^{\infty} f''\, p\, dx \;=\; -\int_{-\infty}^{\infty} f'\, \partial_x p\, dx \;=\; \int_{-\infty}^{\infty} f\, \partial_{xx} p\, dx.$$

Moving the time derivative inside the integral on the left of Equation 3 then gives

$$\int_{-\infty}^{\infty} f(x) \left[\partial_t p(x,t) - \tfrac{1}{2}\, \partial_{xx} p(x,t)\right] dx = 0.$$

This holds for every test function f, so by the fundamental lemma of the calculus of variations the bracket must vanish:

$$\partial_t p(x,t) = \tfrac{1}{2}\, \partial_{xx} p(x,t). \tag{4}$$

This is the FPE for Brownian motion: the heat equation with diffusion coefficient $D = 1/2$. Its solution starting from $W_0 = 0$ is the Gaussian density with variance t.

Strictly, the argument shows that p satisfies Equation 4 in the weak sense, that is, when integrated against every test function. It holds pointwise wherever p is smooth enough, as it is here for all $t > 0$.

2.2 A general SDE

Now let X_t solve

$$dX_t = \mu(X_t, t)\,dt + \sigma(X_t, t)\,dW_t, \tag{5}$$

and let $p(x,t)$ be the density of X_t. Itô's formula gives $df(X_t) = \left[\mu f' + \tfrac{1}{2}\sigma^2 f''\right] dt + \sigma f'\,dW_t$, and the same steps lead to

$$\frac{d}{dt}\int f\,p\,dx = \int \left[\mu\,f' + \tfrac{1}{2}\,\sigma^2 f''\right] p\,dx.$$

Integrating by parts once in the first term and twice in the second moves every derivative onto the products μp and $\sigma^2 p$, and the fundamental lemma gives the general FPE:

$$\partial_t p(x,t) = -\partial_x\left[\mu(x,t)\,p(x,t)\right] + \tfrac{1}{2}\,\partial_{xx}\left[\sigma^2(x,t)\,p(x,t)\right]. \tag{6}$$

The placement of the coefficients matters. The drift multiplies the density before the first derivative is taken; the squared diffusion coefficient multiplies the density before the second derivative is taken, with a factor of one half. Equivalently, $\partial_t p = \mathcal{L}^* p$, where $\mathcal{L} f = \mu f' + \tfrac{1}{2}\sigma^2 f''$ is the generator of the process and $\mathcal{L}^*$ is its adjoint.

2.3 Geometric Brownian motion

For $dX_t = \mu X_t\,dt + \sigma X_t\,dW_t$ with constant μ and σ, Equation 6 becomes

$$\partial_t p = -\mu\,\partial_x(x\,p) + \tfrac{1}{2}\,\sigma^2\,\partial_{xx}(x^2 p). \tag{7}$$

2.4 The Ornstein–Uhlenbeck process

For $dX_t = -\kappa X_t\,dt + \sigma\,dW_t$ with $\kappa > 0$, Equation 6 becomes

$$\partial_t p = \kappa\,\partial_x(x\,p) + \tfrac{1}{2}\,\sigma^2\,\partial_{xx} p. \tag{8}$$

3 Worked solutions

3.1 Ornstein–Uhlenbeck

Starting from $X_0 = x_0$, the solution of Equation 8 is Gaussian,

$$X_t \sim N\!\left(x_0\,e^{-\kappa t},\ \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa t}\right)\right),$$

as can be checked by substituting this density into Equation 8. As $t \to \infty$ it converges to the stationary law $N(0, \sigma^2/2\kappa)$. The stationary law can also be found directly. Setting $\partial_t p = 0$ in Equation 8 and integrating once gives $\kappa x\,p + \tfrac{1}{2}\sigma^2 p' = c$ for a constant c, and $c = 0$ because p and p' vanish at $\pm\infty$. Hence $p(x) \propto \exp(-\kappa x^2/\sigma^2)$, the Gaussian density with variance $\sigma^2/2\kappa$.

3.2 Geometric Brownian motion

Itô's formula applied to $\log X_t$ gives $\log X_t = \log x_0 + (\mu - \frac{1}{2}\sigma^2)\,t + \sigma W_t$, so X_t is lognormal:

$$p(x,t) = \frac{1}{x\,\sigma\sqrt{2\pi t}}\,\exp\left(-\frac{\left[\log(x/x_0) - (\mu - \frac{1}{2}\sigma^2)\,t\right]^2}{2\sigma^2 t}\right), \qquad x > 0,$$

which satisfies Equation 7. Unlike the other two examples, geometric Brownian motion has no stationary distribution: its density keeps spreading out.

3.3 Cox–Ingersoll–Ross

The Cox–Ingersoll–Ross (CIR) process,

$$dv_t = \kappa(\theta - v_t)\,dt + \sigma\sqrt{v_t}\,dW_t, \qquad \kappa, \theta, \sigma > 0,$$

models short-term interest rates (Cox et al. 1985) and the variance in Heston's stochastic volatility model. Its stationary density $\varphi(v)$ on $(0,\infty)$ solves the stationary version of Equation 6,

$$0 = -\frac{d}{dv}\left[\kappa(\theta - v)\,\varphi\right] + \frac{1}{2}\,\frac{d^2}{dv^2}\left[\sigma^2 v\,\varphi\right].$$

Integrating once shows that the probability flux $J(v) = \kappa(\theta - v)\varphi - \frac{1}{2}\frac{d}{dv}[\sigma^2 v\varphi]$ is constant in v. For a stationary distribution, with no probability flowing in or out at the boundaries, that constant is zero. Expanding the derivative with the product rule,

$$\kappa(\theta - v)\,\varphi = \tfrac{1}{2}\sigma^2\,(\varphi + v\,\varphi') \qquad \Longrightarrow \qquad \frac{\varphi'}{\varphi} = \frac{2\kappa\theta/\sigma^2 - 1}{v} - \frac{2\kappa}{\sigma^2},$$

so $\varphi(v) \propto v^{\alpha-1}e^{-\beta v}$ with

$$\alpha = \frac{2\kappa\theta}{\sigma^2}, \qquad \beta = \frac{2\kappa}{\sigma^2}. \tag{9}$$

The stationary law is therefore Gamma with shape α and rate β, with mean θ and variance $\theta\sigma^2/2\kappa$.

The shape parameter carries the most important structural fact about the process. $\alpha \geq 1$ exactly when $2\kappa\theta \geq \sigma^2$, which is the Feller condition: the condition under which the process never reaches zero (Feller 1951). When it holds, the stationary density is bounded near zero; when it fails, the density diverges at zero (while remaining integrable), and the process touches zero from time to time. Both the -1 in the exponent and the factor of 2 in α and β come from placing the coefficients correctly in Equation 6: the -1 from the product rule, and the 2 from the factor of one half.

3.4 A simulation check

Figure 1 compares the three solutions with simulations of the SDEs themselves. The simulations use the Euler–Maruyama scheme, which needs only the SDEs and none of the formulas derived above.

The moments agree as well. The simulated Ornstein–Uhlenbeck variance at $t = 10$ is 0.504, against the stationary value 0.5. The simulated mean and variance of geometric Brownian motion at $t = 1$ are 1.107 and 0.116, against $e^{\mu} \approx 1.105$ and $e^{2\mu}(e^{\sigma^2} - 1) \approx 0.115$. The simulated CIR mean and variance are 0.0403 and 0.000626, against $\theta = 0.04$ and $\theta\sigma^2/2\kappa = 0.000625$.

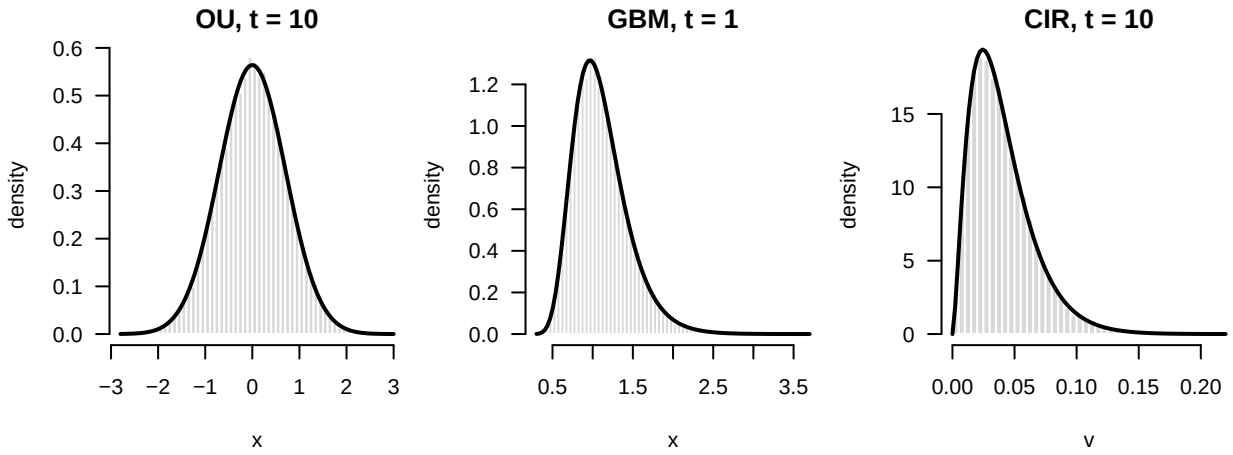


Figure 1: Derived densities (lines) against Euler–Maruyama simulations of the SDEs (histograms, 20,000 paths each). Left: Ornstein–Uhlenbeck with $\kappa = 1$, $\sigma = 1$, started at $x_0 = 2$, at $t = 10$. Middle: geometric Brownian motion with $\mu = 0.1$, $\sigma = 0.3$, $x_0 = 1$, at $t = 1$. Right: Cox–Ingersoll–Ross with $\kappa = 2$, $\theta = 0.04$, $\sigma = 0.25$, at $t = 10$.

4 Numerical and learning-based methods

Closed-form solutions like those above are the exception. Most FPEs have to be solved numerically, and the available methods fall into two broad families.

4.1 Grid-based methods

Finite-difference and finite-element methods discretize the state space (and time) and replace the derivatives in Equation 6 with algebraic approximations, turning the equation into a large, sparse linear system. In one to three dimensions they are accurate, well understood, and fast. A good scheme also has to respect what makes p a density: it should stay non-negative, conserve total probability, and treat boundaries, such as the zero boundary of the CIR process, with care. Pichler et al. (2013) compare finite-difference and finite-element solvers for the FPE and emphasize the main limitation they share: the number of grid points, and with it the computational cost, grows very quickly with the dimension of the state.

4.2 Learning-based methods

A second family represents the solution with a neural network. A physics-informed neural network (Raissi et al. 2019) is trained so that the residual of the differential equation is small at many sampled points, with extra terms in the loss for boundary and initial conditions. The network's derivatives are computed by automatic differentiation, so no mesh is needed. Karniadakis et al. (2021) review this approach and its extensions, including its ability to combine noisy data with a known equation. A related but distinct line of work, Han et al. (2018), rewrites high-dimensional parabolic equations, including Black–Scholes-type and HJB equations, as backward SDEs and uses neural networks to approximate the gradient of the solution. It is not a physics-informed network, but it shares the aim of avoiding a grid.

Several papers adapt these ideas to the FPE specifically. Xu et al. (2020) train a network on the equation's residual and add the normalization condition $\int p\,dx = 1$ to the loss, so that the network cannot settle on the trivial solution $p = 0$, with penalty weights chosen to avoid poor local optima. Hu et al. (2025) target very high-dimensional FPEs: they first learn the score, the gradient of the log-density, by score matching or a physics-informed variant, and then recover the log-density itself by integrating an ordinary differential equation driven by the learned score. Liu et al. (2021) reverse the direction of the problem. Starting from a sequence of observed, noisy densities, they recover the drift and diffusion terms that generated them, building those terms into the network as trainable weights and denoising the densities as they fit.

For low-dimensional problems such as the examples in Section 3, a grid-based solver is usually simpler and more accurate. The case for learning-based methods rests on high-dimensional problems, where grids are infeasible, and on problems that combine an equation with data.

5 Conclusion

The Fokker–Planck equation turns a question about random paths into a deterministic equation for their density. Its derivation needs only Itô's formula and integration by parts. Its solutions for the Ornstein–Uhlenbeck, geometric Brownian motion, and Cox–Ingersoll–Ross processes are explicit, and the CIR case shows how a structural property of a process, whether it can reach zero, appears directly in its stationary density. In finance the equation recovers local volatility from option prices; in economics it describes how distributions across heterogeneous agents evolve. Where closed forms run out, grid-based methods handle low-dimensional problems well, and learning-based methods extend the equation's reach to high dimensions and to problems that combine equations with data.

Revision notes

This note was written in May 2024 and revised in September 2026. The derivation in Section 2 was corrected: the 2024 version misstated the support condition on the test function, dropped a minus sign in the first integration by parts (it reappeared in the next step, so the final equation was right), omitted "$= 0$" from one equation, and attributed the vanishing of the stochastic term to $\mathbb{E}[dW_t] = 0$ rather than to the mean-zero property of the Itô integral. Equations that the 2024 version called solutions of the FPE are now labeled as the equations they are, and the worked solutions in Section 3, including the CIR stationary law and the simulation check, are new.

The background now treats the SDE and Fokker–Planck descriptions as equivalent rather than competing, and credits Kolmogorov. The applications were made concrete with Dupire's formula and the HJB–Kolmogorov forward system. Section 4 was rewritten, and its attributions were corrected: the high-dimensional BSDE method is now credited to Han et al. (2018) and distinguished from physics-informed networks. The reference list was corrected and completed.

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